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Enhancement of the spectral selectivity of complex samples by measuring them in a frozen state at low temperatures in order to improve accuracy for quantitative analysis. Part II. Determination of viscosity for lube base oils using Raman spectroscopy†

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The use of selectivity-enhanced Raman spectra of lube base oil (LBO) samples achieved by the spectral collection under frozen conditions at low temperatures was effective for improving accuracy for the determination of the kinematic viscosity at 40 °C (KV@40). A collection of Raman spectra from samples cooled around −160 °C provided the most accurate measurement of KV@40. Components of the LBO samples were mainly long-chain hydrocarbons with molecular structures that were deformable when these were frozen, and the different structural deformabilities of the components enhanced spectral selectivity among the samples. To study the structural variation of components according to the change of sample temperature from cryogenic to ambient condition, *n*-heptadecane and pristane (2,6,10,14-tetramethylpentadecane) were selected as representative components of LBO samples, and their temperature-induced spectral features as well as the corresponding spectral loadings were investigated. A two-dimensional (2D) correlation analysis was also employed to explain the origin for the improved accuracy. The asynchronous 2D correlation pattern was simplest at the optimal temperature, indicating the occurrence of distinct and selective spectral variations, which enabled the variation of KV@40 of LBO samples to be more accurately assessed.

Introduction

The spectral acquisition of complex mixture samples under frozen conditions at low temperatures effectively enhanced Raman spectral selectivity among samples and improved the accuracy of their compositional analysis as previously demonstrated in part I.[1] This study was a basic proof-of-concept analytical investigation that employed synthetic hydrocarbon mixtures of similar molecular structure and, further, it was necessary to evaluate this methodology for analysis of real samples to demonstrate its practical analytical usage. For this purpose, we selected lube base oils (LBOs), highly complex mixtures containing predominantly long-chain (C_{16}–C_{25}) *n*-paraffins and iso-paraffins, and attempted to find an optimal sample temperature for Raman spectral collection leading to accurate determination of their kinematic viscosity at 40 °C (KV@40). KV@40, the viscosity measured at 40 °C, is one of the most important properties governing the quality of LBOs. It is a bulk physical property resulting from the contributions of individual components as well as their intermolecular interactions. Therefore, the better spectral discrimination among individual components in LBO samples should be helpful to more accurately correlate with the variation of KV@40, and an exploration of measurement conditions that provide enhanced spectral selectivity was of great interest. This effort was more challenging since the compositional complexity of LBOs is much greater than that of the 11-component mixture used in part I of this study.[1]

Initially, *n*-heptadecane and pristane (2,6,10,14-tetramethylpentadecane) were selected as model molecules to represent the *n*-paraffin and iso-paraffin components of LBOs, respectively. Raman spectra of these two compounds were continuously collected while varying the sample temperature from cryogenic to ambient temperature, and their spectral variations were examined. Similarly, Raman spectra of LBO samples were periodically collected during the same temperature variation (a total of 90 snapshots). Using the spectral datasets collected at each snapshot, partial least squares (PLS)[2,3] was used to determine the KV@40 of samples, and an optimal snapshot yielding the most accurate analysis was determined. To explain the improved accuracy, the PLS loading feature at the optimal snapshot was compared with those at other snapshots. To derive an alternative supportive justification, a two-dimensional

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(2D) correlation analysis[4–7] able to detail spectral variations of all LBO samples at each snapshot was employed, and the resulting 2D correlation patterns were examined.

Experimental

Samples and Raman spectral collection

n-Heptadecane and pristane were purchased from Sigma-Aldrich. Twenty-eight LBO samples from diverse international manufacturers were provided by a petrochemical company in Korea. The samples were cautiously prepared with the aid of an expert to include wider variations of KV@40. KV@40s of samples were determined using the ASTM D445 method. Sample colors varied from transparent to light yellow.

Raman spectra were collected using a wide area illumination (WAI) scheme (PhAT system, Kaiser Optical Inc., Ann Arbor, MI, USA) as previously described.[8] A glass vial (diameter: 9.8 mm) containing the LBO sample (volume: 2 mL) was initially immersed in liquid nitrogen for 5 minutes to decrease the sample temperature to cryogenic temperature. Then the vial was removed and immediately positioned at the focal point of the WAI scheme for Raman spectral acquisition. As the temperature of the samples naturally increased to near 0 °C, Raman spectra were continuously collected at 5 second intervals, and a total of 90 spectra (or snapshots) were obtained for each sample. Fig. 1 shows the actual temperature variation of samples, where the *x*-axis is designated as both the snapshot number and elapsed time. The error bars correspond to standard deviations of the temperature measurements from all 28 samples. Spectra for each sample were collected in triplicate. Every LBO sample was divided into three different glass vials, and Raman spectra were separately collected for these divided samples. The resolution of collected Raman spectra was 4 cm^{-1}. All spectral pre-processing and multivariate methods, including baseline correction, normalization, PLS, and 2D correlation analysis, were performed using Matlab Version 7.0 (Math Works Inc., MA, USA).

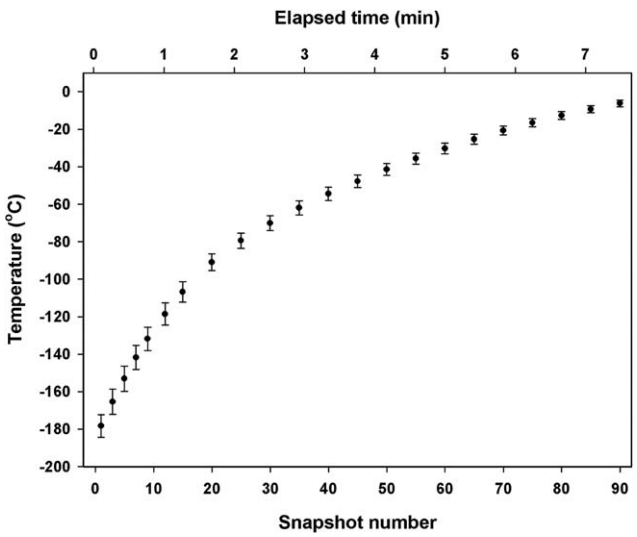

Fig. 1 Actual temperature variation of LBO samples during Raman spectral collection, where the *x*-axis designates both snapshot number and elapsed time. The error bars indicate the corresponding standard deviations of individual temperature measurements for all 28 LBO samples.

Results and discussion

Raman spectral variation of *n*-heptadecane and pristane by temperature change

Fig. 2(a) shows normalized Raman spectra (1500–690 cm^{-1} range) of *n*-heptadecane collected during the temperature variation from cryogenic to near room temperature. For normalization, the baselines of raw spectra were linearly corrected at 1500, 1200, 987, and 690 cm^{-1}, the area under the 1500–690 cm^{-1} range was calculated, and each baseline-corrected spectrum was divided by the corresponding area for normalization. Only the 1st, 5th, 10th, 20th, 30th, 50th, 70th, and 90th snapshots out of 90 total snapshots are displayed for clarity. Since the melting temperature of *n*-heptadecane is 20 °C, all of the spectra shown in the figure were collected when it was in the solid state. For comparison, the spectrum of liquid *n*-heptadecane (thick black line) collected at 25 °C is also shown. For detailed examination, spectral features in the 1480–1400, 1315–1290, 1145–1058, and 940–820 cm^{-1} ranges are highlighted in Fig. 2(b)–(e), respectively. In these figures, the intensity scale of the liquid *n*-heptadecane spectrum (right *y*-axis) was properly adjusted to maximize the comparison.

As shown in Fig. 2(a), the spectral features of *n*-heptadecane dramatically differed depending on whether it was a solid or a liquid. The bands were more distinct and narrower for the frozen sample. Especially in the 1480–1400 cm^{-1} range (see Fig. 2(b)), four distinct bands were apparent for solid *n*-heptadecane while only two broadly overlapped bands were observed with liquid *n*-heptadecane. The broad 1302 cm^{-1} band acquired in the liquid state moved to a lower wavenumber around 1296 cm^{-1} and became narrower for solid *n*-heptadecane as shown in Fig. 2(c). The shapes of bands from the two phases in the 1145–1058 cm^{-1} range also greatly differed as shown in Fig. 2(d) while the positions of the corresponding bands were relatively close to each other. Contrary to the spectral differences observed in Fig. 2(b), the spectral features in the 940–820 cm^{-1} range became simpler for the frozen sample as shown in Fig. 2(e). Three broad peaks became one narrow peak upon the phase change from liquid to solid. All of the spectral variations shown here obviously originate from varied molecular vibrations of *n*-heptadecane depending on its phase as well as the sample temperature.

Although *n*-heptadecane maintained a solid form from the 1st to the 90th snapshot, diverse spectral variations such as shifts of the band position either to a higher or lower wavenumber, variations of bandwidth, and minor intensity changes were observed. The shift of the band position was most noticeable for the bands shown in Fig. 2(b)–(d). These results indicate that sensitive variations in the molecular vibrations of *n*-heptadecane occur depending on the degree of condensation.

Fig. 3(a) shows Raman spectra (1500–690 cm^{-1} range) of pristane collected during the temperature variation from

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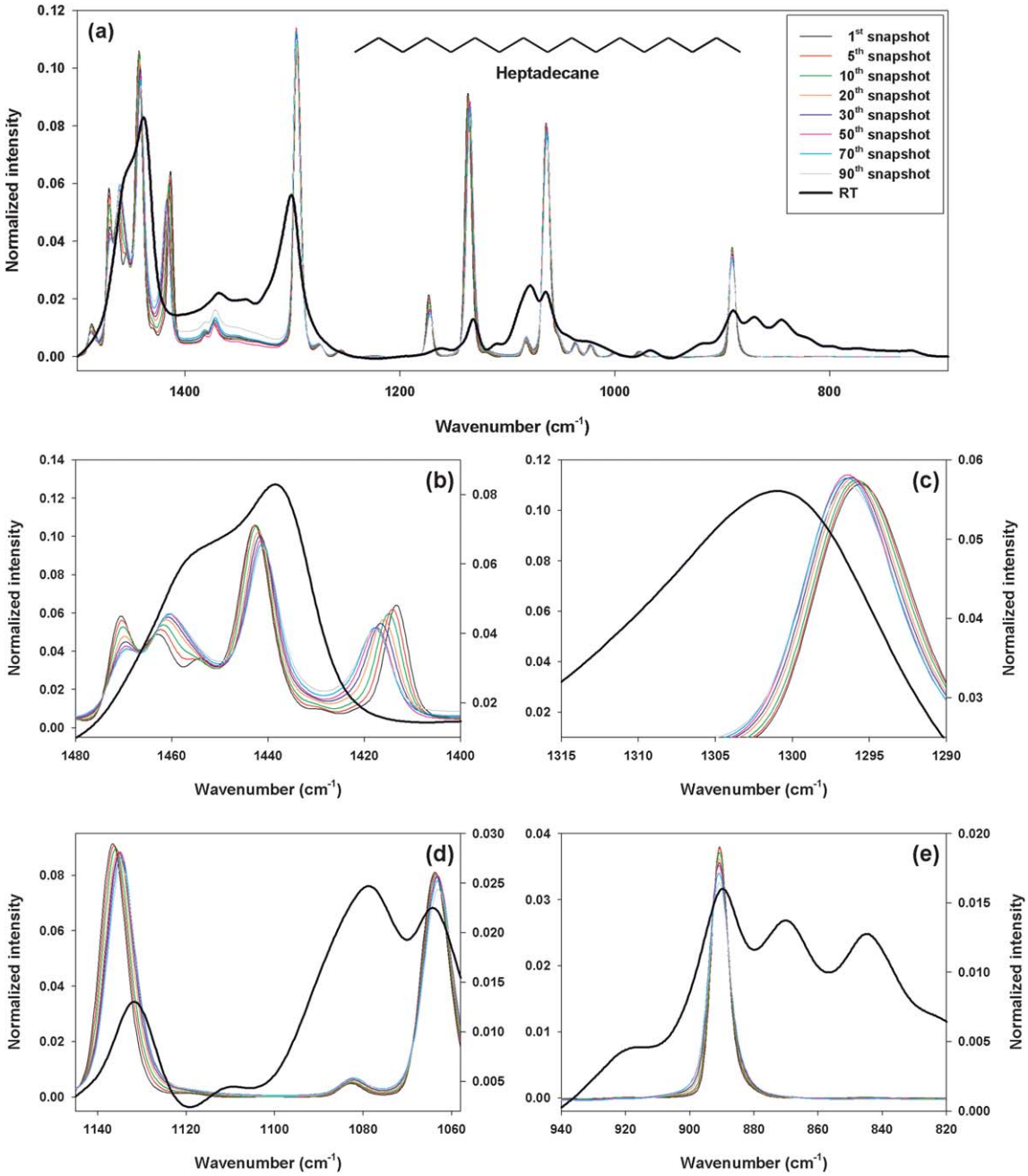

Fig. 2 Normalized Raman spectra of *n*-heptadecane collected at 8 different snapshots and room temperature in the 1500–690 cm^{-1} range (a). For a detailed comparison, the 1480–1400 (b), 1315–1290 (c), 1145–1058 (d), and 940–820 cm^{-1} (e) ranges are highlighted.

cryogenic to near room temperature. The molecular structure of pristane is also shown. Normalized spectra collected at the 1st, 5th, 10th, 20th, 30th, 50th, 70th, and 90th snapshots and at room temperature (thick black line) are displayed. For normalization, the baselines of raw spectra were linearly corrected at 1500, 1222, 1003, and 690 cm^{-1}, and each baseline-corrected spectrum was divided by the corresponding area under the 1500–690 cm^{-1} range. For a detailed examination, the spectral features in the 1400–1290, 1180–1140, 1078–1004, and 896–700 cm^{-1} ranges are highlighted in Fig. 3(b)–(e), respectively. The melting temperature of pristane is −100 °C, which approximately corresponds to the 20th snapshot. The abrupt intensity drop after the 10th snapshot shown in Fig. 3(c)–(e) is due to the phase change of the sample from solid to liquid. Unlike *n*-heptadecane, the spectral features of pristane acquired either in the solid or liquid states were relatively similar. No significant changes in band shape were observed during the temperature change, although some minor variations in bandwidth and band shape were observed.

Overall, the molecular structure of *n*-heptadecane is more easily deformable in a frozen state, such that the corresponding temperature-induced spectral variation is substantial as was

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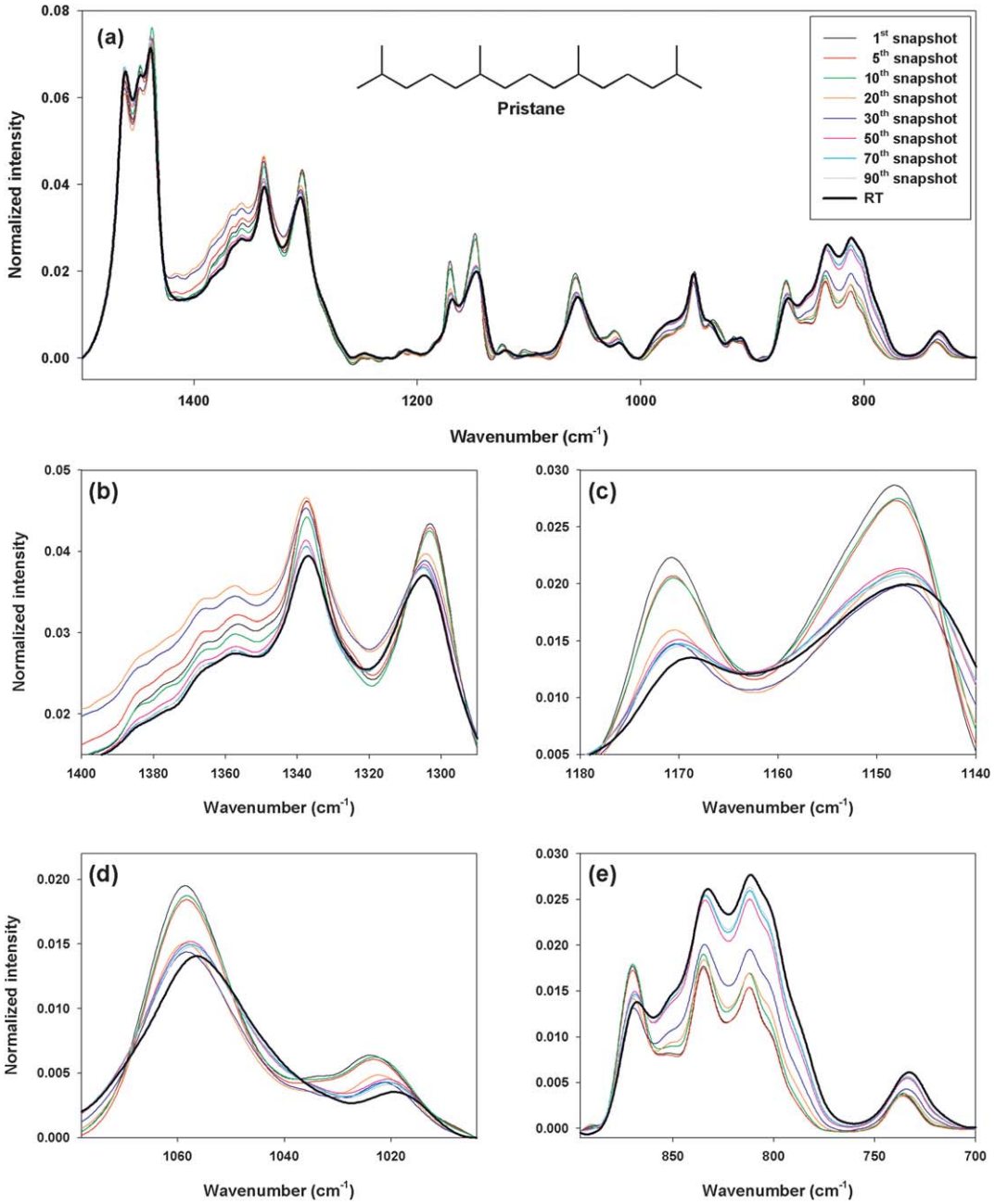

Fig. 3 Normalized Raman spectra of pristane collected at 8 different snapshots and room temperature in the 1500–690 cm^{-1} range (a). For a detailed comparison, the 1400–1290 (b), 1180–1140 (c), 1078–1004 (d), and 896–700 cm^{-1} (e) ranges are highlighted.

previously observed for the short-chain linear hydrocarbons examined in part I.[1] By contrast, the molecular structure of pristane is less deformable in a frozen state due to the presence of four methyl branches along the backbone, a rationale for the relatively marginal spectral changes observed with the temperature variation. LBO samples consist of large numbers of various hydrocarbons; therefore, the structural deformability of individual components would be dissimilar and the corresponding induced spectral variations would be different under frozen conditions. Therefore, a particular condition that provides maximum spectral selectivity among a group of components could exist.

Raman spectral variation of LBO samples according to temperature change

Fig. 4(a) shows Raman spectra (1500–690 cm^{-1} range) of a LBO sample (KV@40: 16.43 cSt) collected during the temperature variation. Only eight normalized spectra acquired at the 1st, 5th, 10th, 20th, 30th, 50th, 70th, and 90th snapshots are shown for comparison. For normalization, the baselines of raw spectra were linearly corrected at 1500, 1410, 1217, 996, and 690 cm^{-1}, and each baseline-corrected spectrum was divided by the corresponding area under the 1500–690 cm^{-1} range. For a detailed examination of spectral variations, the 1470–1430, 1320–1280 and 1180–1040 cm^{-1} ranges

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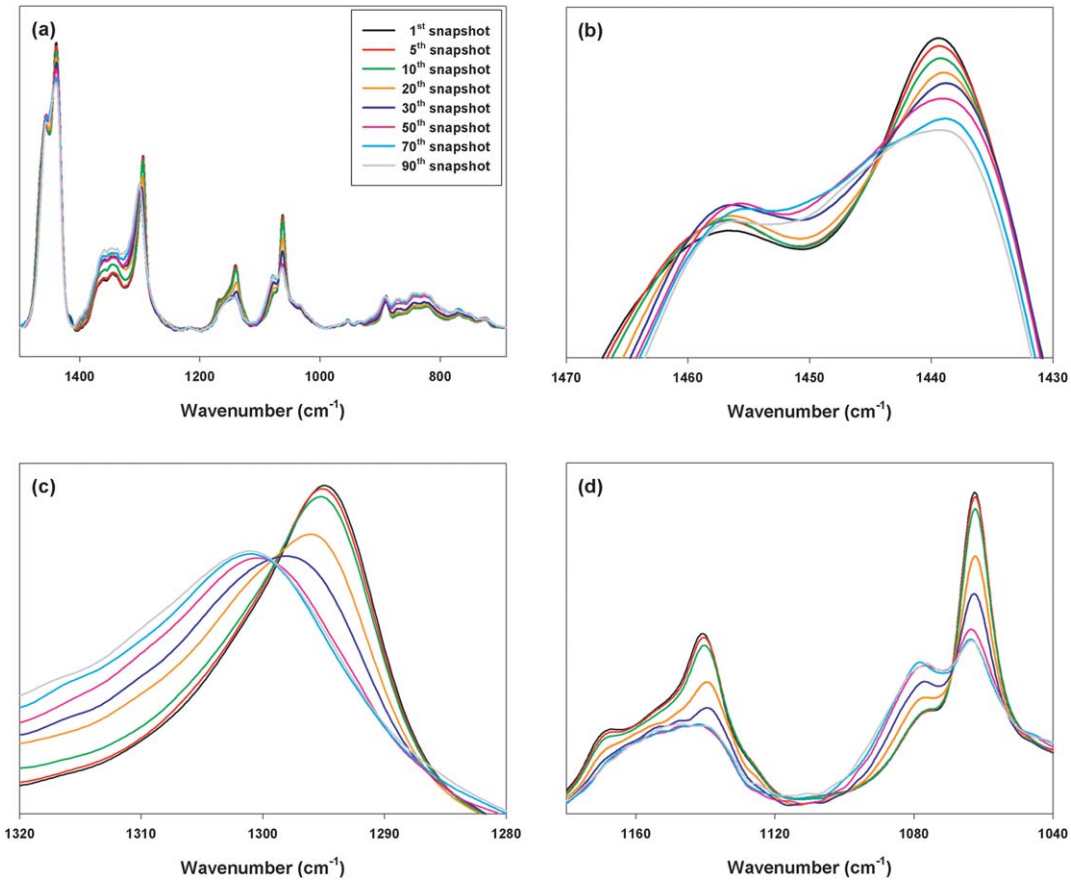

Fig. 4 Normalized Raman spectra of a LBO sample (KV@40: 18.6 cSt) collected at 8 different snapshots in the 1500–690 (a), 1470–1430 (b), 1320–1280 (c), and 1170–1040 cm^{-1} (d) ranges.

are magnified in Fig. 4(b)–(d), respectively. Relevant vibrational assignments for the bands of LBOs are summarized in Table 1. The melting temperatures of LBO samples approximately ranged from −60 °C to −40 °C, corresponding to the 40th to 50th snapshot, depending on their composition.

Table 1 Vibrational assignments for the bands of LBO samples[9–11]

Band position (cm^{-1})	Vibrational assignment
1457	CH_3 asymmetric bending
1440	CH_2 scissoring in alkanes
1350	CH_2 scissoring
1302	CH_2 bending of alkanes
1167	Skeletal stretching primarily in iso-alkanes
1140	Skeletal stretching primarily in *n*-alkanes
1080	Skeletal stretching of alkanes and alkenes
1060	Stretching primarily in *n*-alkanes
1035	C–C skeletal stretching
975	CH wag in dialkyl ethylenes
955	C–C skeletal stretching
934	C–C skeletal stretching
890	Cyclopentane ring breathing
870	Alkylated cyclohexane ring breathing
844	Alkylated cyclohexane ring breathing
831	Symmetric C–C–C stretch
773	Symmetric C–C–C stretch, C–C stretch in C–C$(CH_3)_2$

As shown in Fig. 4(b), changes in band shape were apparent upon temperature elevation. The intensity of the 1440 cm^{-1} band (CH_2 scissoring) decreased with the increase of temperature. In addition, as the temperature was elevated, the 1456 cm^{-1} band (CH_3 asymmetric bending) became broader and its position somewhat varied. From the 50th snapshot, spectral features became less characteristic, such as a broad overlapping of these two bands due to the phase change from solid to liquid. As shown in Fig. 4(c), the intensity of the 1295 cm^{-1} band (CH_2 bending) in the 1st snapshot decreased, and its position gradually shifted to a higher wavenumber as the temperature increased. After the phase change from solid to liquid, this band remained around 1302 cm^{-1} but broadened. The intensities of the major bands in the 1180–1040 cm^{-1} range (see Fig. 4(d)) became weaker and less distinct as the temperature was elevated. In particular, the distinct spectral features in the 1180–1120 cm^{-1} range from the frozen state nearly disappeared after the sample thawed, as observed in spectra collected after the 50th snapshot. All of the spectral variations shown above resulted from complexly altered molecular vibrations of individual components in the LBO sample induced by the temperature perturbation. The spectral variations were not as dramatic as those observed in the temperature-varied spectra of *n*-heptadecane due to the considerable spectral overlap with bands from other components; however, selectivity-enhanced

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features were still apparent in the spectra and could be a valuable source for a more accurate description of LBO properties.

Determination of KV@40 at each snapshot using PLS

PLS was used to determine the KV@40 of LBO samples using the datasets collected at each snapshot. The 1500–690 cm^{-1} range, shown in Fig. 4(a), was used for PLS. Errors were obtained by a full cross-validation method, with 21 and 7 samples randomly assigned to calibration and validation sets, respectively. The standard error of calibration (SEC) and standard error of prediction (SEP) were obtained for all possible combinations for sample assignment combinations. Finally, the mean standard error of calibration (MSEC) and mean standard error of prediction (MSEP) were calculated by averaging the individual SEC and SEP values.

Fig. 5 shows the MSEP results from the 90 individual snapshots acquired during the temperature variation. In all cases, 4 factors were used for each PLS model. The MSEPs from the first ten snapshots were substantially lower than those from the following snapshots. After the 10th snapshot, the MSEP increased rather sharply until the 30th snapshot and then oscillated without an improvement in accuracy. The lowest MSEP was obtained at the 7th snapshot (around −160 °C), and the improvement of accuracy was statistically significant based on a *t*-test at the 95% confidence level compared to those from later snapshots such as the 30th snapshot. This result implies that Raman spectral features of LBO samples under frozen states at a low temperature are more descriptive to follow the variation of KV@40. The improved accuracy appears to be reasonable since the spectral features of the two pure components as well as LBOs were much sharper in the earlier snapshots as shown in Fig. 2–4.

To explain the improved accuracy, the loadings employed for each PLS model were initially examined since the weighted spectral information at each snapshot to describe the variation of KV@40 could be compared to each other. The first loadings used for each measurement in the 1320–1260, 1217–1100 and 1100–1000 cm^{-1} ranges are shown in Fig. 6(a)–(c), respectively. Only the loadings used in the 1st, 5th, 10th, 20th, 30th, 50th, 70th, and 90th snapshots are shown for clarity. As shown in Fig. 6(a), the weight on the 1292 cm^{-1} band (CH_2 bending) was strongest for the 1st snapshot and gradually became weaker for the following snapshots. Around the 30th snapshot, the weight of this band was quite small, and no distinct loading features were observed after the 50th snapshot. This observation substantiates the degraded accuracy starting

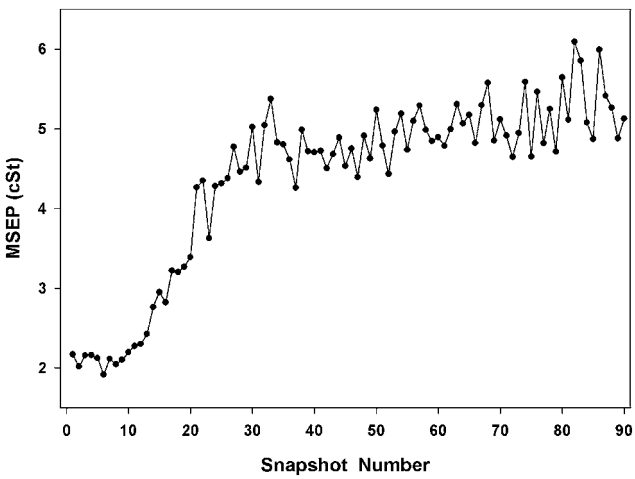

Fig. 5 MSEPs for the determination of KV@40 from the 1st to 90th snapshots.

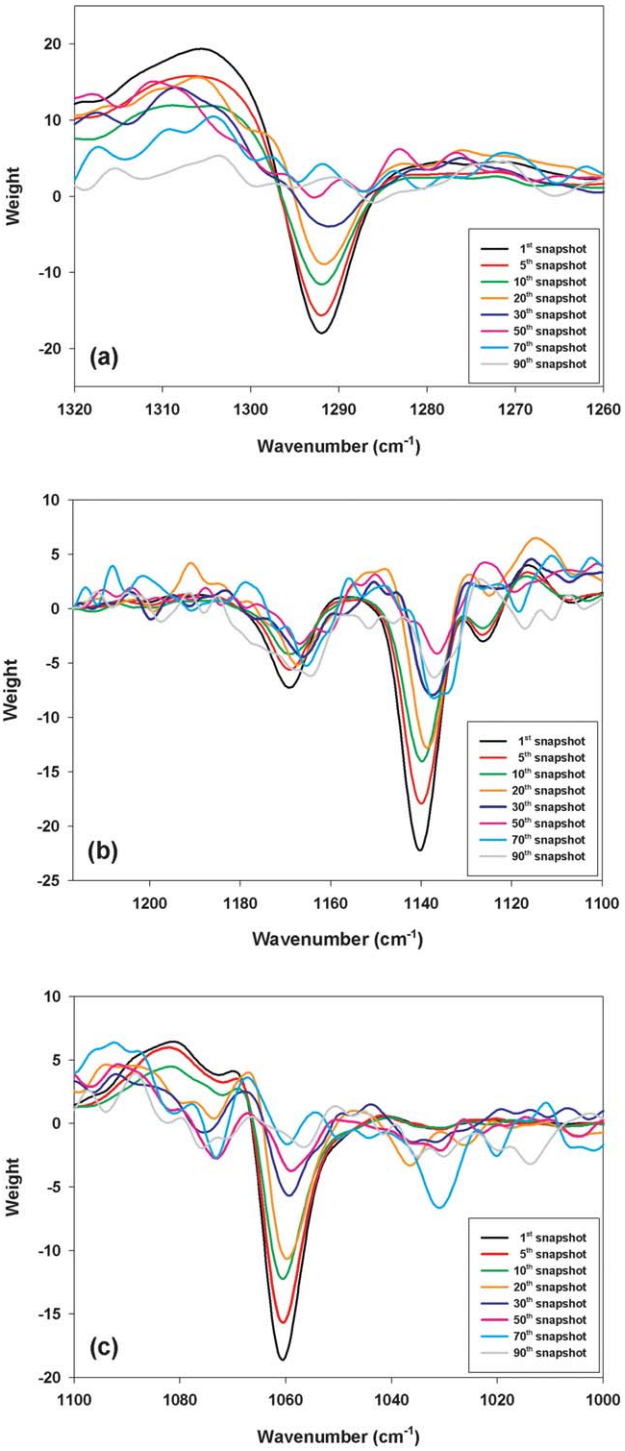

Fig. 6 The first PLS loadings in the 1320–1260 (a), 1217–1100 (b), and 1100–1000 cm^{-1} (c) ranges. Only loadings used in the 1st, 5th, 10th, 20th, 30th, 50th, 70th, and 90th snapshots are shown for clarity.

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around the 30th snapshot as well as the large fluctuation of MSEP from the 30th to 90th snapshots, as shown in Fig. 5.

A similar trend was observed for loadings in the 1217–1100 cm^{-1} and 1100–1000 cm^{-1} ranges as shown in Fig. 6(b) and (c), respectively. Weighted bands at 1140 and 1060 cm^{-1} correspond primarily to the asymmetric and symmetric C–C stretching of all-*trans* chains, respectively.[12–14] This means that the stretching of the all-*trans* chain in LBO samples becomes more dominant with the decrease of sample temperature. The weighted positions of these two bands remained constant until the 10th snapshot and slightly shifted to a lower wavenumber at the 20th snapshot. This implies that some discrete structural variation occurs between the 10th and 20th snapshot, although it would be difficult to specifically explain the observed band shift. As shown in Fig. 6, PLS models developed from the earlier snapshots (such as the first ten snapshots) effectively weighted the enhanced spectral features that were more descriptive of the variation of KV@40.

Generalized two-dimensional (2D) correlation analysis

Although the enhanced spectral variation among LBO samples between the 1st and 10th snapshots was confirmed, it would be of interest to address the best accuracy at the 7th snapshot. In addition, the steep degradation of accuracy from the 10th to 20th snapshot was quite surprising since the corresponding loading features were still distinct, as shown in Fig. 6. To provide a perspective on the above-mentioned inquiries, a generalized 2D correlation analysis,[4–7] which is able to provide a detailed qualitative evaluation of spectral variations under a given external perturbation, was employed. The spectral datasets collected at each snapshot were individually subject to a 2D correlation analysis; therefore, the spectral variation present in each dataset can be regarded as a result of perturbation induced by a change of KV@40 at a given temperature.

Fig. 7 shows the asynchronous 2D correlation maps acquired from 15 different snapshots (1st, 3rd, 5th, 7th, 9th, 12th, 15th, 20th, 25th, 30th, 35th, 40th, 50th, 70th, and 90th), using the same 1500–690 cm^{-1} range. Red and blue colors indicate positive and negative correlations, respectively. The asynchronous correlation was primarily investigated since it could effectively recognize minute spectral variations. Many small positive and negative correlation peaks were present in the 2D maps, and the complexity of the asynchronous correlation varied substantially. Correlation patterns acquired from the 5th to 9th snapshots were considerably simpler than those acquired from other snapshots, and the correlation was simplest at the 7th snapshot.

The simple asynchronous correlation implies that distinct spectral variations occurred according to the change of KV@40. By contrast, for a snapshot yielding a complicated asynchronous correlation, the quite complex (or less distinct) spectral variation was used to model the property variation, and thus the resulting PLS accuracy would be degraded. The simplest asynchronous correlation pattern at the 7th snapshot matches the best accuracy in the KV@40 measurement. The asynchronous

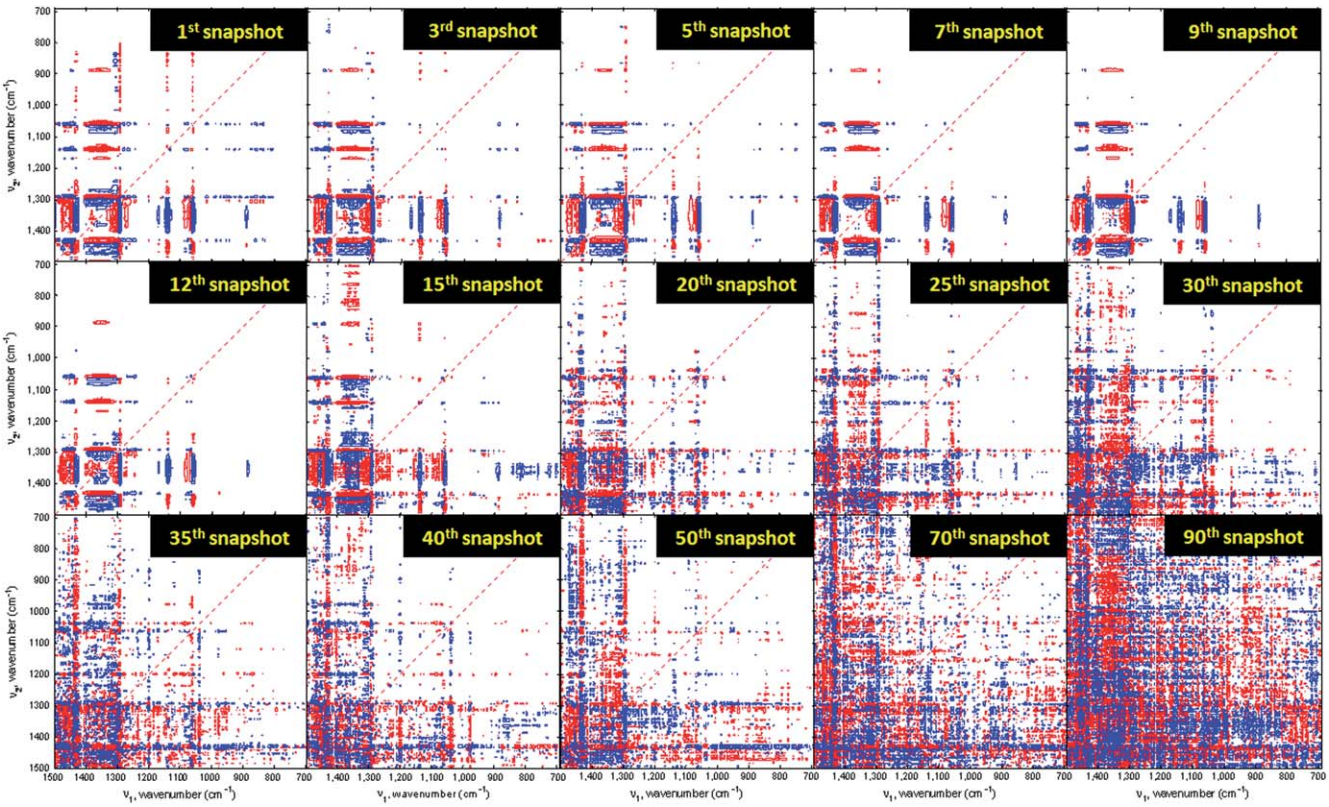

Fig. 7 Asynchronous 2D correlation maps obtained using datasets collected at 15 different snapshots. Red and blue colors indicate positive and negative cross peaks, respectively.

correlation patterns from the 12th to 20th snapshots became complex rather suddenly, which indicates an initiation of complicated spectral variations among the samples and provides an explanation for the sharp increase of the prediction error over this period. Overall, the degree of complexity in the asynchronous 2D correlation was quite supportive of the rationale for the variation of MSEP shown in Fig. 5.

Conclusion

The collection of Raman spectra of LBO samples under frozen states at low temperatures, such as around −160 °C, improved the accuracy of the KV@40 determination. The dissimilar structural deformability of LBO components at low temperatures was the major source of spectral selectivity enhancement. In addition, the 2D asynchronous correlation analysis was helpful in describing the variation of measurement accuracy according to the sample temperature. We hope that this study encourages analysts to make an effort to find optimal conditions to improve the spectral selectivity among samples rather than simply perform spectral collection at room temperature, especially when sample spectra are highly overlapped and the use of multivariate methods is inevitable for calibration. The demonstrated strategy should also be effective for the analysis of other complex samples, such as various petroleum-derived samples (*e.g.* gasoline, kerosene, diesel) and edible oils. Obviously, optimized conditions would vary depending on the characteristics of samples such as compositional complexity and molecular structures.

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